

EXP 9

Aim

To create, compile, deploy, and test an ERC-20 token using Solidity in Remix IDE and deploy the token on the Sepolia Ethereum test network using MetaMask.

Theory

An **ERC-20 token** is a fungible token standard on the Ethereum blockchain. Fungible means that every unit of the token is identical and has the same value as every other unit of the same token. ERC-20 defines a common interface that allows wallets, exchanges, and smart contracts to interact with tokens in a standard way.

The main ERC-20 functions include `totalSupply()`, `balanceOf()`, `transfer()`, `approve()`, `allowance()`, and `transferFrom()`. Important events include `Transfer` and `Approval`.

In this experiment, an ERC-20 token named **EXP09Token** with the symbol **EXP** was created using OpenZeppelin's ERC-20 implementation. An initial supply of **1000 EXP** was minted to the deploying account. The contract was compiled and deployed on the **Sepolia testnet** through MetaMask. The deployed contract functions were then executed to verify the token's properties, supply, and balances.

Program

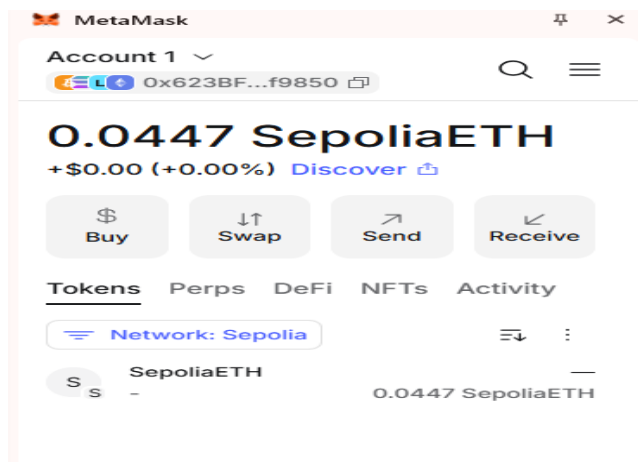
```
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.20;
```

```
import {ERC20} from "@openzeppelin/contracts/token/ERC20/ERC20.sol";
```

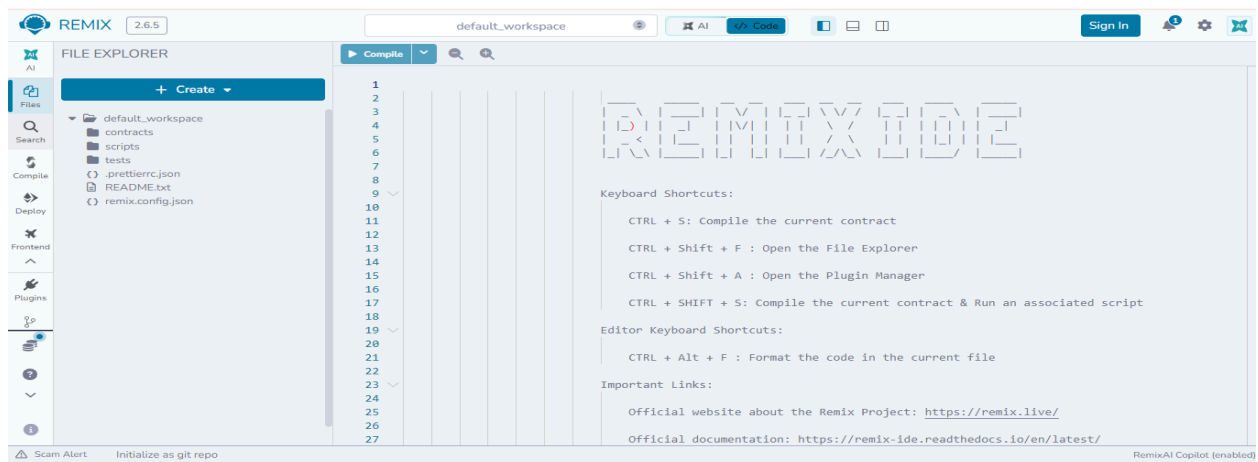
```
contract ChiragToken is ERC20 {
    constructor(uint256 initialSupply)
        ERC20("EXP09 Token", "EXP")
    {
        _mint(msg.sender, initialSupply * 10 ** decimals());
    }
}
```

Output

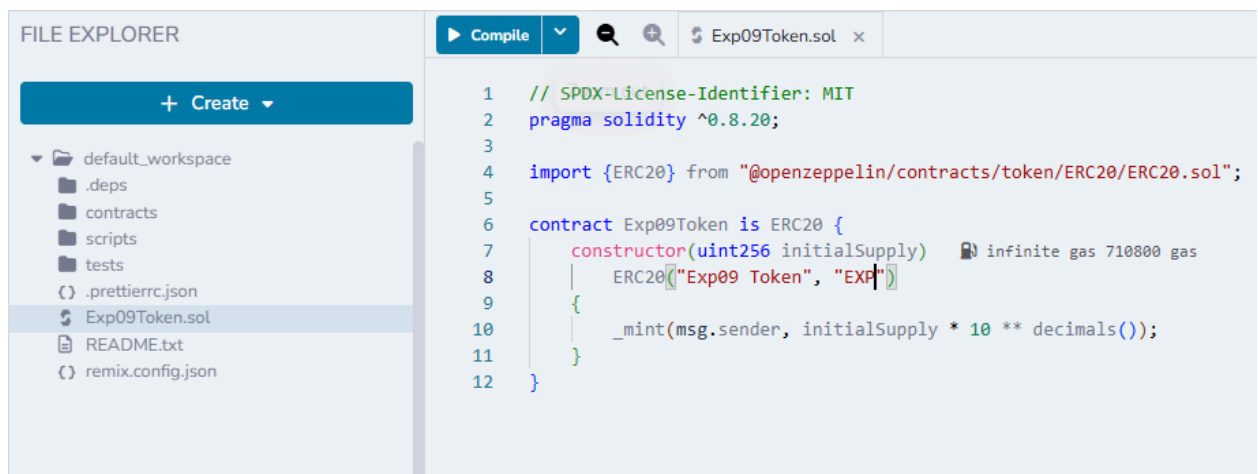
1. MetaMask Wallet and Sepolia Test ETH



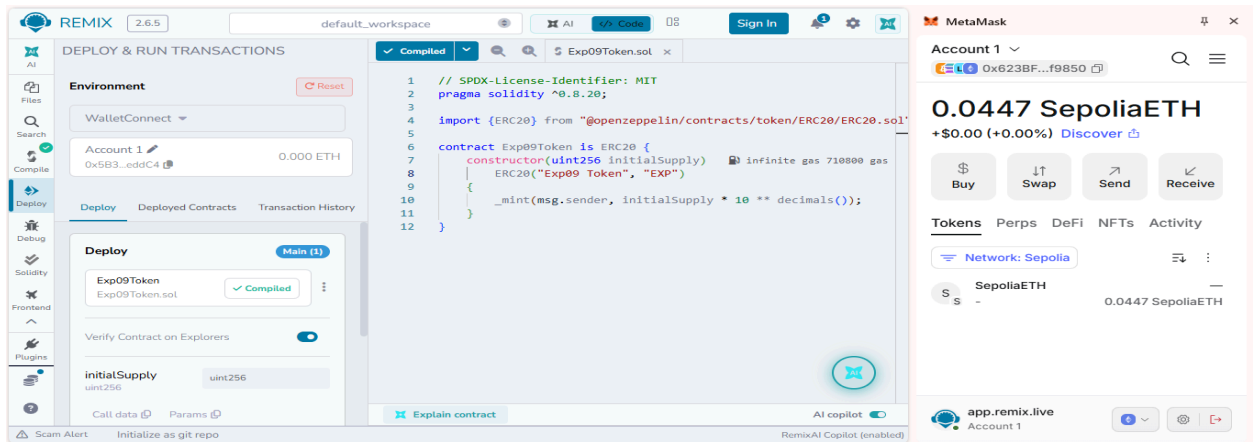
2. Remix IDE



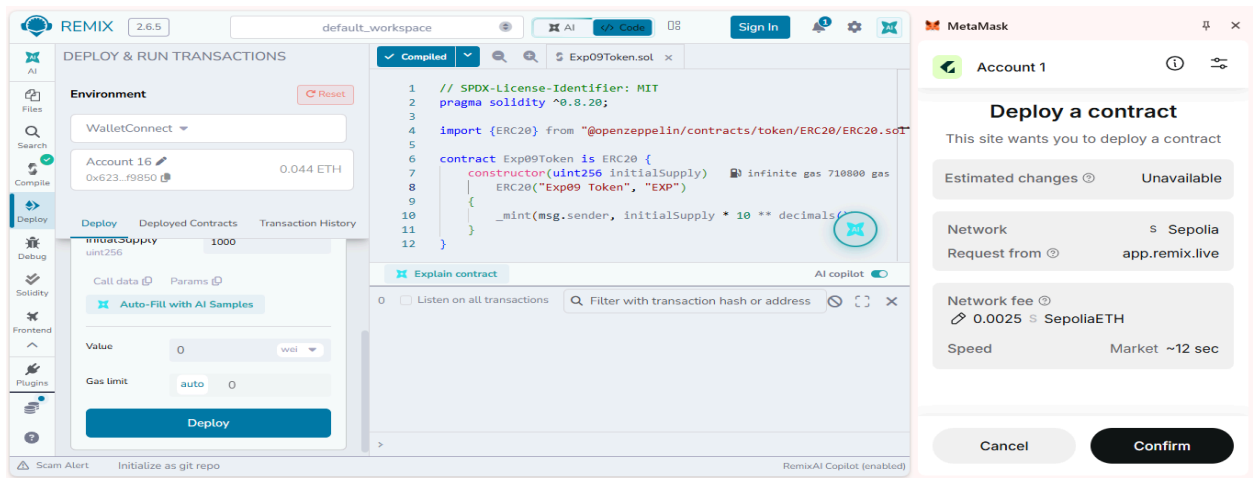
3. Solidity ERC-20 Smart Contract



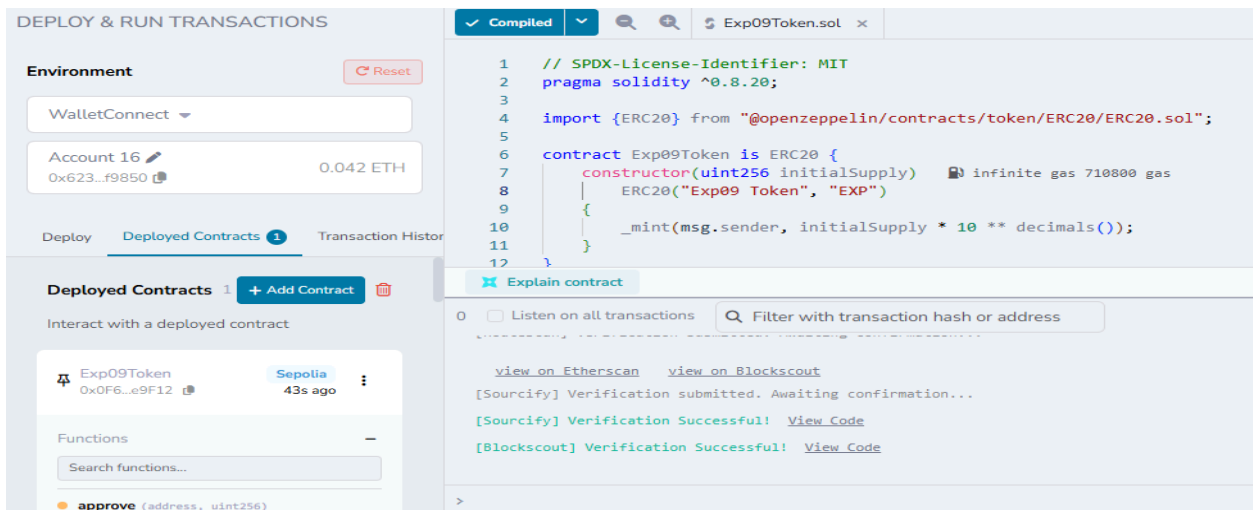
4. Contract Compilation



5. MetaMask Wallet Connection



6. Contract Deployment



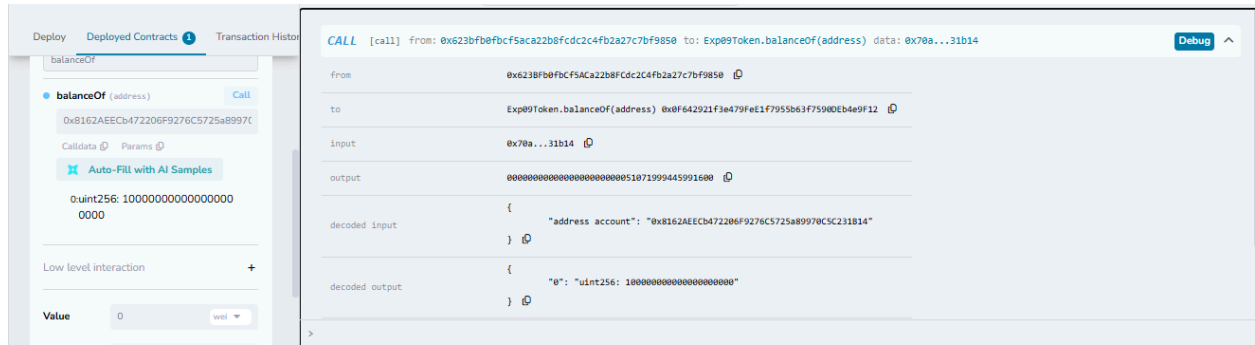
7. ERC-20 Function Execution

The following functions were tested:

- `name()` — verified the token name as **EXP09 Token**
- `symbol()` — verified the token symbol as **EXP**
- `decimals()` — verified the decimal precision
- `totalSupply()` — verified the total token supply
- `balanceOf()` — verified the token balance of the account
- `transfer()` — tested transfer of tokens
- Other available ERC-20 functions were also examined/executed as part of the experiment.

Results of the executed ERC-20 contract functions.

[illegible]



8. Sepolia Blockchain Transaction Verification

TRANSACTION ACTION
0x623Bfb0f...7c7bf9850 Transfer 100 EXP to 0x8162AEEC...C5C231B14
EIP-7702: 0x623Bfb0f...7c7bf9850 Delegate to 0x63c0c19a...A07DAE32B

[This is a Sepolia Testnet transaction only]

Transaction Hash: 0xcd4e79e1ffacd686d66541a5ba29880ff5e33b7cd3c1d24832f088001a23ff5e

Status: Success

Block: 11846982 10 Block Confirmations

Timestamp: 2 mins ago Oct-05-2026 05:57:36 AM +UTC

From: 0x623Bfb0fbCf5AcA22b8FcDc2C4fb2a27c7bf9850

Interacted With (To): 0xdb9B1e94B5b69Df7e401DDbedE43491141047dB3

ERC-20 Tokens Transferred: All Transfers Net Transfers

From 0x623Bfb0f...7c7bf9850 To 0x8162AEEC...C5C231B14 For 100 ERC-20: Exp09 Token (EXP)

Value: 0 ETH

Transaction Fee: 0.0005110643168068 ETH

Gas Price: 2.452323977 Gwei (0.000000002452323977 ETH)

Conclusion

The ERC-20 token was successfully created using Solidity and OpenZeppelin, compiled and deployed through Remix IDE on the Sepolia test network. The token's standard functions were successfully tested, demonstrating the creation, management, and transfer of a fungible blockchain token.